

Einstein-VED Theory: Solving the Mysteries of Quantum Mechanics

Complementary Clarifications to the Einstein-VED Theory

This document provides additional explanations and expansions to the original Einstein-VED theory, emphasizing geometric interpretations, proper time, and effects related to mass.

It serves as a complement to the original work documented on Zenodo:

[The Theory That Will Reconcile Quantum Mechanics with Einstein's Relativity \(PDF in Spanish\)](#)

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Introduction

The Einstein-VED theory proposes a geometric and relativistic interpretation of quantum phenomena, based on the principles of Einstein's relativity and on a **global wavefront** shared by all particles involved.

Unlike quantum mechanics (QM), Einstein-VED does not postulate fundamental probabilities nor wavefunction collapse: the observed indeterminacy is merely a consequence of projecting the complete structure of the system onto the space-time parameters of a limited observer.

Shared Proper Time and Correlations

- All particles from a wavefront generated at t_0 share a common **proper time** τ
- This proper time does not depend on the observer's time or space

- Correlations between entangled particles emerge **naturally and deterministically** from the coherence of the wavefront

Implications

1. **No superluminal communication:** correlations do not require information transmission between separated particles
2. **No local hidden variables:** the correlation arises from the global structure of the wavefront, not from hidden parameters
3. **The apparent non-locality of quantum mechanics** is explained as the projection of a continuous geometric reality onto the observer's reference frames

Critique of Quantum Probability

Quantum mechanics interprets the wavefunction as a fundamental probabilistic object. Einstein-VED clarifies:

- **Classical probability:** quantifies what we do not know
- **Quantum probability:** is applied even when the state is completely defined (Ψ known), generating an epistemological contradiction
- **Einstein-VED solution:** the observed indeterminacy is simply the observer's lack of information about the complete structure in τ
- The wavefunction collapse is not a physical process: it is an update of the observer's knowledge reflecting coherence with the global geometry

Quantum Correlations Without Collapse or Signaling

Consider a particle pair with outcomes $A, B \in \{+1, -1\}$ and measurement orientations a, b :

$$P(A, B \mid a, b, \tau) = \frac{1}{4} (1 + C(a, b, \tau)AB)$$

- Marginalizing over B or A yields:

$$P_A(A \mid a) = \sum_B P(A, B \mid a, b) = \frac{1}{2}, \quad P_B(B \mid b) = \frac{1}{2}$$

- The observed correlations are:

$$E(a, b) = \langle AB \rangle = \int d\mu(\tau) C(a, b, \tau)$$

If we choose $C(a, b, \tau) = -\cos(\alpha - \beta)$, we obtain exactly the singlet state correlation, **without violating causality or introducing collapse**.

Conceptual Advantages of Einstein-VED

1. Naturally explains quantum correlations that are "magic" for quantum mechanics
2. Eliminates the need for hidden variables and collapse
3. Maintains relativistic covariance: predictions do not depend on the observer's space-time foliation
4. Allows for establishing falsifiable predictions: deviations from correlations can arise if the shared proper time is manipulated or the wavefront coherence is altered

Conclusion

Einstein-VED provides a deterministic and geometric interpretation of quantum phenomena, respecting relativity and eliminating conceptual contradictions of quantum mechanics.

The theory transforms apparent indeterminacy into **observational ignorance**, and turns entanglement and non-locality into **natural consequences of the geometric continuity** of physical systems.

Expansion of Proper Time and Mass in the Einstein-VED Theory

Context

Within the Einstein-VED theory, all particles from a wavefront generated at t_0 share a common proper time τ .

This naturally explains quantum correlations without resorting to hidden variables or wavefunction collapse.

A new aspect arises when considering the **presence of mass and particle velocity**.

Velocity Limit and Proper Time

- For moving particles, the following holds:

$$v = \frac{ds}{dt}$$

- In the relativistic limit $v \rightarrow c = 1$ (in our units), the differential of proper time tends to zero:

$$dt' \rightarrow 0$$

- However, **the presence of mass guarantees** that $v < 1$, ensuring a proper time $\tau > 0$
- Mass also implies acceleration in the environment:

$$\frac{d^2s}{dt^2} \neq 0$$

Expanded Proper Time Field

Due to the finite velocity and the acceleration associated with mass, the **observed proper time "expands"** as a function of velocity and acceleration:

$$d\tau = F(v, a)$$

where:

- $v = ds/dt$ is the particle velocity
- $a = d^2s/dt^2$ is the acceleration in the environment
- F is a **scalar field** defined over the wavefront that describes the local evolution of proper time

This field can be considered as a **new object of study**, connecting mass, motion, and the geometry of the wavefront.

Consequences

1. **No absolute time:** proper time depends locally on the velocity and acceleration of each particle
2. **Emergent relativistic effects:** time dilation and other phenomena associated with velocity and mass are naturally derived from the wavefront geometry
3. **Influence on correlations:** particles with mass and acceleration modify the propagation of their wavefront, which can dynamically affect correlations without the need for superluminal interaction
4. **Continuous observational field:** this approach allows describing the evolution of proper time and its effects on entangled systems in a deterministic and geometric manner, without resorting to ontological probabilities

Future Directions

The study of $F(v, a)$ opens possibilities such as:

- Quantitatively deriving how acceleration affects wavefront coherence
- Establishing predictions about deviations in quantum correlations for accelerated particles
- Exploring connections with relativistic and cosmological phenomena from a unified framework

This development naturally integrates **mass, velocity, and acceleration** into the Einstein-VED theory, extending its scope beyond quantum correlations.